Hemodynamic Attenuation: Pruning Attention Heads the Way the Brain Prunes Vessels

Anonymous submission

Abstract

The brain keeps a blood vessel only while the flow it carries is worth its metabolic upkeep. Vessels that neighbours can cover regress gradually, and new ones sprout where demand returns. We turn this principle into a rule that sizes a transformer's attention during training. Each head gets a valve, and its value is the loss that would remain if it closed and the other heads re-fit to cover for it, the analogue of collateral flow. A head closes, gradually, when its value falls below a fixed upkeep cost. No target size is set. The rule is greedy minimisation of loss plus upkeep, and this makes testable predictions. On a retrieval task whose minimal number of heads $k^\star$ is known, the rule kept exactly $k^\star$ heads in 29 of 30 runs and never broke the model. Three published pruning methods each broke it in every run. The number kept did not depend on the cost until the cost crossed the value of one needed head, where theory predicts a collapse to one head. The collapse occurred there in 36 of 36 runs. On two-layer induction the rule recovered the known previous-token and induction heads in 10 of 10 runs. On GPT-2 small it kept 17 of 144 heads for indirect object identification, 15 of them in the published circuit. Under random head failure it kept the number of backup heads predicted before the runs. When the task's demand rose and fell during training, adding sprouting, the analogue of angiogenesis, let the rule regrow the missing heads in 10 of 10 runs and close them again when demand fell.

Introduction

A trained transformer usually has more attention heads than its task needs (Michel, Levy, and Neubig 2019; Voita et al. 2019), and trained models keep backup heads that take over when others are removed (Wang et al. 2023; McGrath et al. 2023). Existing pruning methods fix the final size by hand, as a target sparsity (Zhu and Gupta 2017), a penalty strength (Voita et al. 2019) or a threshold (Conmy et al. 2023), and most prune after training, in one step.

The cerebral circulation solves a similar allocation problem without a target size. Astrocytes and the cells of the neurovascular unit sense local demand and adjust vessel diameter (Iadecola 2017). Vessels that carry little flow regress gradually rather than being cut (Korn and Augustin 2015), and tissue survives the loss of a vessel when collateral vessels supply its territory. Murray (1926) showed that vessel size follows from minimising the work of moving blood plus the metabolic cost of keeping it. A vessel is kept only while it earns its upkeep. Developing neural circuits likewise overproduce and then prune connections (Navlakha, Barth, and Bar-Joseph 2015).

We carry this principle over to attention heads and call the result *hemodynamic attenuation*. Our contributions are:

- A rule that values each head by the loss that remains after the other heads compensate for it, charges a fixed upkeep per open head, and closes heads gradually during training, with reopening. It is greedy minimisation of loss plus upkeep (Proposition 1).
- A benchmark whose minimal circuit is known in size and roles, and a prediction of the cost range in which the rule recovers it, confirmed in 36 of 36 runs.
- Evidence that the rule recovers the minimal count without breaking the model, recovers the induction circuit, finds most of the published indirect-object circuit in GPT-2 small, keeps a predicted number of backups under damage, and, with sprouting, regrows heads when demand returns and closes them when it falls.

Hemodynamic Attenuation

Valves. Each head's output passes through a valve $g_h \in [0, 1]$ before the heads are summed, $\hat{T} = \sum_h g_h \, O_h W_O^h$. A head with $g_h = 0$ adds nothing and receives no gradient, so a model with k open valves is exactly a k-head model.

Collateral value. For the set S of open heads, let $L^*(S) = \min_{W_O} L$ be the loss after the output weights of the heads in S are re-fitted, with their attention held fixed. The collateral value of an open head is $v_h = L^*(S \setminus h) - L^*(S)$, and the gain of a closed head is $L^*(S) - L^*(S \cup h)$. Both come in closed form from one inverse of the heads' output covariance on a batch of cached outputs. The score is the Optimal Brain Surgeon score (Hassibi and Stork 1993) applied to whole heads, as in ZipLM (Kurtic, Frantar, and Alistarh 2023). What is new is its use. The re-fit is the analogue of collateral flow and makes a copy worthless. If two heads output the same signal, the survivor takes over its twin's weight and the loss does not change.

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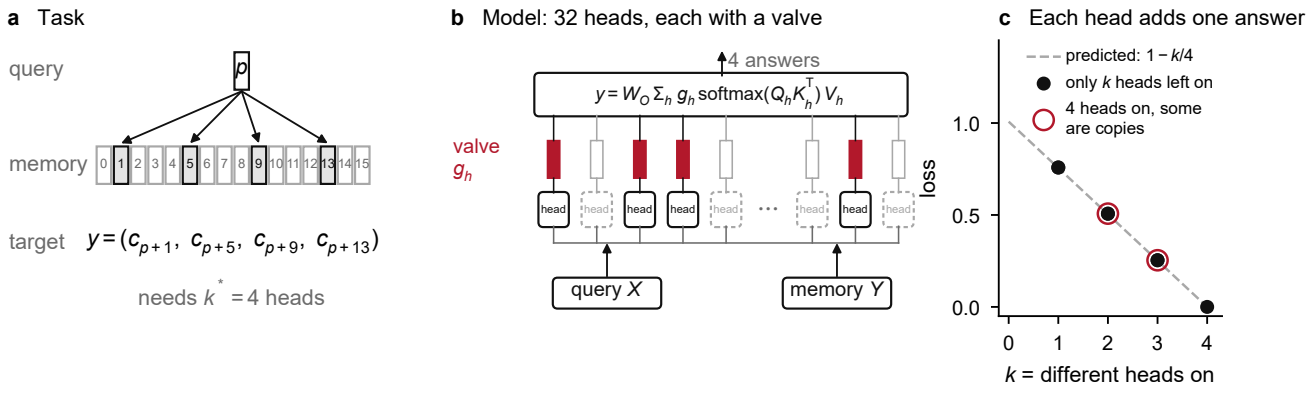


Figure 1: The retrieval benchmark. **(a)** A query names a position p, and the target is the items at R fixed offsets from it, here $R = 4$. **(b)** One cross-attention layer with 32 heads, each behind a valve g_h. The rule decides which valves stay open (red). **(c)** Loss of a trained model with only k of its heads left on (black) follows the rank law $1 - k/R$ (dashed). Leaving 4 heads on, some of them copies of each other, gives the loss of the number of *different* heads (red), so a copy is worth nothing. Losses in units of L_{triv}.

Schedule. All valves start open. After a warm-up of 10% of training, every 25 steps the rule (1) computes all values, (2) closes the lowest-valued open head if $v_h < \pi$, and otherwise (3) reopens the highest-gain closed head if its gain exceeds 2π. A closing valve falls by $1/100$ per step, so a head regresses over 100 steps while the others adapt. The upkeep is $\pi = 0.03\, L_{\text{triv}}$, where L_{triv} is the loss of predicting the mean.

The rule optimises an objective. Let $J(S) = L^*(S) + \pi|S|$, the loss plus the upkeep of the open heads, which is the form of Murray's cost.

Proposition 1. *Closing h lowers J exactly when $v_h < \pi$, and reopening h lowers J by more than π. At a fixed point of the rule, no single removal lowers J, and no single addition lowers it by more than π. If needed heads are worth more than π and redundant heads less, the fixed point keeps the needed heads and only those, for any such π.*

The first two claims follow from $J(S \setminus h) - J(S) = v_h - \pi$ and $J(S \cup h) - J(S) = \pi - \text{gain}_h$. The rule is therefore forward–backward greedy selection (Zhang 2011) over groups of weights, run while the heads themselves are still learning. The fade and the warm-up handle the fact that L^* changes as the heads train. The proposition makes a falsifiable prediction. The number of heads kept should not depend on π while π lies between the value of a redundant head and the value of a needed one, and it should change abruptly when π leaves that range.

A Benchmark with a Known Minimal Circuit

A memory of $N = 16$ slots holds random items c_j, redrawn for every example. A query names a position p, and the target is the R items at fixed offsets, $(c_{p+\delta_1}, \ldots, c_{p+\delta_R})$, slots taken modulo N (Figure 1a). A head returns one weighted average of the items per query, which is one linear combination of the R unknowns, so recovering R items needs R heads with different attention patterns, one per offset. With $k \le R$ such heads the lowest achievable loss is

$$L^*(k) = (1 - k/R)\, L_{\text{triv}}, \tag{1}$$

which we confirmed on trained models to within $0.003\, L_{\text{triv}}$ (Figure 1c). A copy of a head adds nothing. Each needed head is therefore worth L_{triv}/R and each redundant head 0, so Proposition 1 predicts exactly $k^\star = R$ heads for any $\pi < L_{\text{triv}}/R$, and a collapse to a single head for $\pi > L_{\text{triv}}/R$, where every needed head costs more than it saves. The model is one cross-attention layer with $H = 32$ heads (Figure 1b), width 128, head width 32 and no MLP. The task is solved when the loss is below $0.02\, L_{\text{triv}}$ ($R^2 \ge 0.98$). We report *exact count* (open heads over the last 40% of training equal $k^\star$) and *never broke* (once solved, the loss never rises above the bar).

Results

Exact count without breaking the model. The rule kept exactly $k^\star$ heads in 29 of 30 runs and never broke the model in 30 of 30 (Figure 2a,b). We implemented three published methods in the same training runs. Michel et al. (Michel, Levy, and Neubig 2019), given the true count, kept it in 28 of 30 runs. ZipLM-style pruning (Kurtic, Frantar, and Alistarh 2023), which applies the same score once and then fine-tunes, kept it in 24 of 30. Learned L_0 gates (Voita et al. 2019; Louizos, Welling, and Kingma 2018) kept it in 9 of 30. All three broke the model in every run. Removing one part of the rule at a time shows what each part does. Without the fade the model broke in every run, like vessels cut rather than regressed. Without the re-fit, which is the Optimal Brain Damage score (LeCun, Denker, and Solla 1990), the rule kept too many heads (2 of 30 exact) because it valued copies. With a random choice of head it broke the model in 24 of 30 runs. When every head started with an exact twin, the rule kept 4 heads and never both members of a pair in 10 of 10 runs, while the score without re-fit kept 8 to 15 heads.

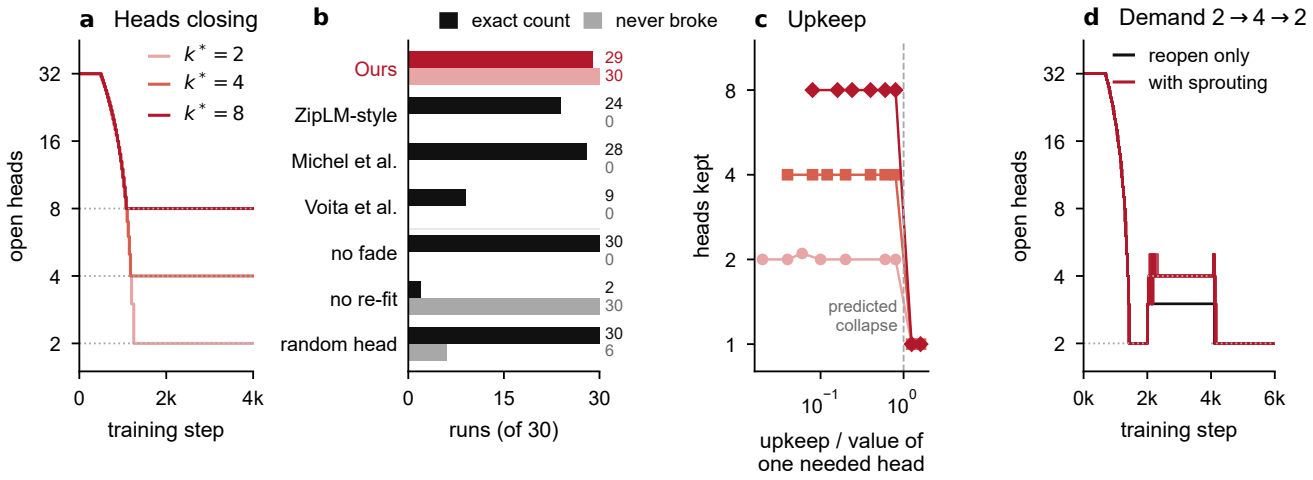


Figure 2: **(a)** Open heads during training, one run per task size, with $k^\star$ dotted. **(b)** Every method on the same 30 runs ($k^\star = 2, 4, 8$, 10 seeds each): runs with exactly $k^\star$ heads (dark) and runs that never broke (light). **(c)** Heads kept against the upkeep π, in units of the value of one needed head, L_{triv}/R. Theory predicts $k^\star$ heads to the left of 1 and one head to the right. **(d)** Demand rises from 2 to 4 offsets and falls back to 2 during training, 10 runs each. Reopening alone (black) stalls at 3 heads in 8 of 10 runs; with sprouting (red) every run regrows to 4 and returns to 2.

The cost does not set the size, until theory says it should. Every upkeep from 0.01 to $0.1\,L_{\text{triv}}$ gave the exact count at $k^\star = 2, 4, 8$. Proposition 1 predicts where this stops. Above $\pi = L_{\text{triv}}/R$ every needed head costs more than it saves, so the rule should close heads down to one. We set π at 0.6, 0.8, 1.25 and 1.6 times L_{triv}/R for $k^\star = 2, 4, 8$, with 3 seeds each. All 18 runs below the breakpoint kept exactly $k^\star$ heads, and all 18 above it kept one (Figure 2c). The final loss of the collapsed runs was 0.50, 0.75 and $0.875\,L_{\text{triv}}$, which is Equation (1) with $k = 1$. The size is set by the task, and the range of costs that recover it is known in advance.

Backup heads under damage. If each head fails with probability p at every step, a head is worth keeping only when it is likely to be needed. Averaging the collateral value over failure patterns gives the expected value of the B-th head, $v(B) = \frac{1-p}{R}\Pr[\text{Bin}(B-1, 1-p) < R]\,L_{\text{triv}}$, and the predicted count is the largest B with $v(B) \geq \pi$. We wrote these predictions down before the runs. They matched in 25 of 30 runs, and the rest were off by one head (Figure 3). Like collaterals, which develop where occlusion is likely, backup heads appear only when failure makes them worth their upkeep.

A second circuit: induction. A two-layer attention-only model with $8 + 8$ heads was trained to continue a repeated sequence of tokens. Its known minimal circuit is a previous-token head in layer 1 and an induction head in layer 2 (Olsson et al. 2022). Cross-entropy has no closed-form re-fit, so here a head is valued by the loss without it. The rule kept $2 + 1$ heads in 10 of 10 runs and never broke the model (Figure 4b), and reopening fired 2 to 7 times per run. The kept heads had the expected roles in every run. Switching off the layer-2 head reduced next-token accuracy on the repeat from 99.9% to 1.8%, chance for 64 tokens (Figure 4a). Layer 1 kept two heads that each missed about 1 in 8 repeated tokens alone and about 1 in 200 together. Trained from scratch, $1 + 1$ heads also solve the task (Figure 4c), so the extra layer-1 head is a kept backup, not a requirement. Michel et al. broke the model in all 10 runs.

A pretrained model: GPT-2 small. To test the rule beyond models we trained, we applied it to the 144 heads of GPT-2 small on indirect object identification ("Then, Mary and John went to the store. John gave a drink to" → "Mary"), whose circuit of 26 heads in seven roles was found by hand (Wang et al. 2023). The weights were frozen, so there is no fade and no re-fit. A closed head is replaced by its mean output on prompts with three different names, as in that work, and a head is valued by the loss without it, as on induction. The loss is the cross-entropy of the correct name, and the upkeep is 0.02 of the gap between the loss with all heads closed and with all heads open. On a frozen model the rule is exactly the greedy minimisation of Proposition 1. It kept 17 heads, and 15 of them are in the published circuit: all three name movers, four backup name movers, three of four S-inhibition heads, and at least one induction, duplicate-token and previous-token head. It kept neither negative name mover, heads that push against the correct name, which a rule that charges upkeep should close. On held-out prompts the 17 heads recovered 88% of the full model's logit difference, against 89% for the published 26-head circuit, with a loss of 0.96 against 1.96 (full model 0.93), and chose the correct name over the subject in 89% of prompts, against 97% for the published circuit. This is a single upkeep and a single set of prompts, and runs at other upkeeps are under way.

Changing demand. Vascular networks also grow where demand returns. We tested this by changing the task during

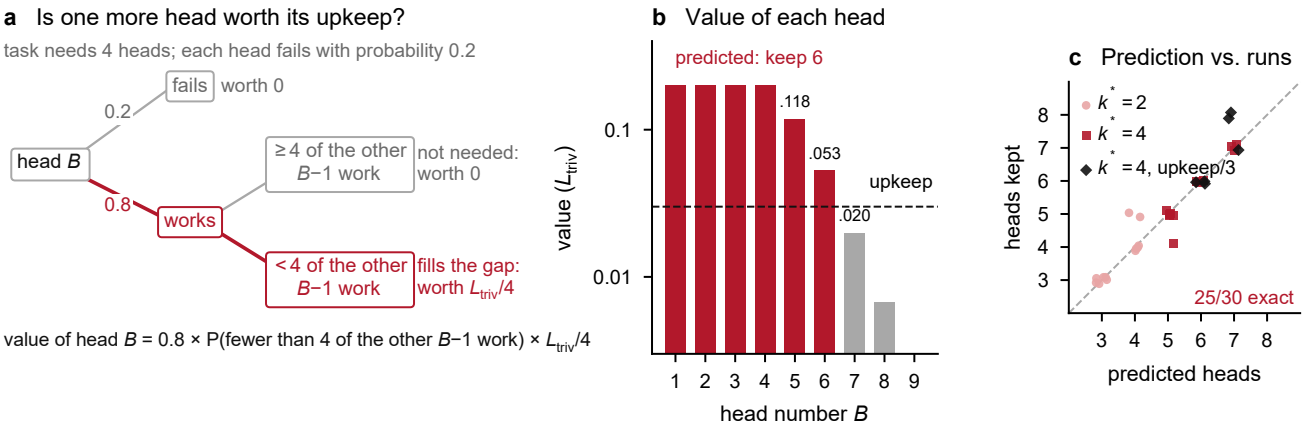


Figure 3: Backup heads, predicted from a failure tree before the runs. **(a)** Each head fails independently with probability p at every step. An extra head earns anything only on one branch: it works, and fewer than $k^\star$ of the other heads work, so it supplies the missing answer, worth $L_{\text{triv}}/k^\star$. Its value is the product along that branch. **(b)** That value for each head when $k^\star = 4$ and $p = 0.2$. The first six beat the upkeep $\pi = 0.03\,L_{\text{triv}}$, so the prediction is six heads (four needed, two backups). **(c)** Heads kept in all 30 damaged runs against the prediction, for p from 0.05 to 0.3 (points jittered). 25 match exactly and the rest are off by one.

training. The run is split into three equal phases of 2,000 steps, in which 2, then 4, then 2 of the four offsets are active, the targets of the inactive offsets being zero. A model with two heads cannot solve the middle phase, since by Equation (1) its loss cannot fall below $0.5\,L_{\text{triv}}$. With reopening alone, only 2 of 10 runs recovered. In the other 8 the rule reopened one head, which learned one new offset, and then stalled at 3 heads with a loss of $0.25\,L_{\text{triv}}$ for the whole phase (Figure 2d). The reason is that a closed head receives no gradient, so it cannot learn the new offset, and its gain stays below 2π. We therefore added *sprouting*, the analogue of angiogenesis driven by hypoxic tissue. If the loss stays above the solved bar for four consecutive checks, the closed head with the highest gain is opened even below 2π, so that it can learn, and the rule closes it again if it does not earn its upkeep. With sprouting, 10 of 10 runs reached exactly 4 heads and solved the task within 100 to 300 steps of the change. When demand fell, all 20 runs closed back to exactly 2 heads within 180 steps.

Size. With exactly $k^\star$ heads, deleting the closed heads leaves 6.3% to 25% of the retrieval model's parameters (12.5% for $k^\star = 4$), and 30.3% of the induction model's. Counting only open heads, training used 0.31 to 0.45 of the dense model's head-steps.

Discussion

The vascular principle that a vessel stays only while it earns its upkeep gives a pruning rule with no target size. It also gives an objective, and the objective predicts where the rule works and where it fails. Three properties come from the biology, and each is needed. The re-fit plays the role of collateral flow and removes copies. Gradual regression keeps the model intact. Reopening and sprouting play the role of regrowth, and sprouting is needed because a closed head, like a regressed vessel, cannot adapt until it is supplied again. The correspondence is a design principle, not a model of cerebral blood flow.

Limitations. The tasks are small and synthetic, chosen so that the minimal circuit is known, and the retrieval model has a single layer. The re-fit is exact only for the output weights. On induction the rule kept one layer-1 head more than the minimum, and a value that re-fits each layer to rebuild what the next layer reads may close it. Closed heads are still computed during training, so the saving in training compute is nominal until they are deleted. On GPT-2 the weights were frozen, so the result tests the value and the stopping rule but not pruning during training, and it rests on one upkeep and one set of prompts. Next we will prune GPT-2 while fine-tuning it, and apply the rule to a cross-attention model of two wearable signals, chest and wrist.

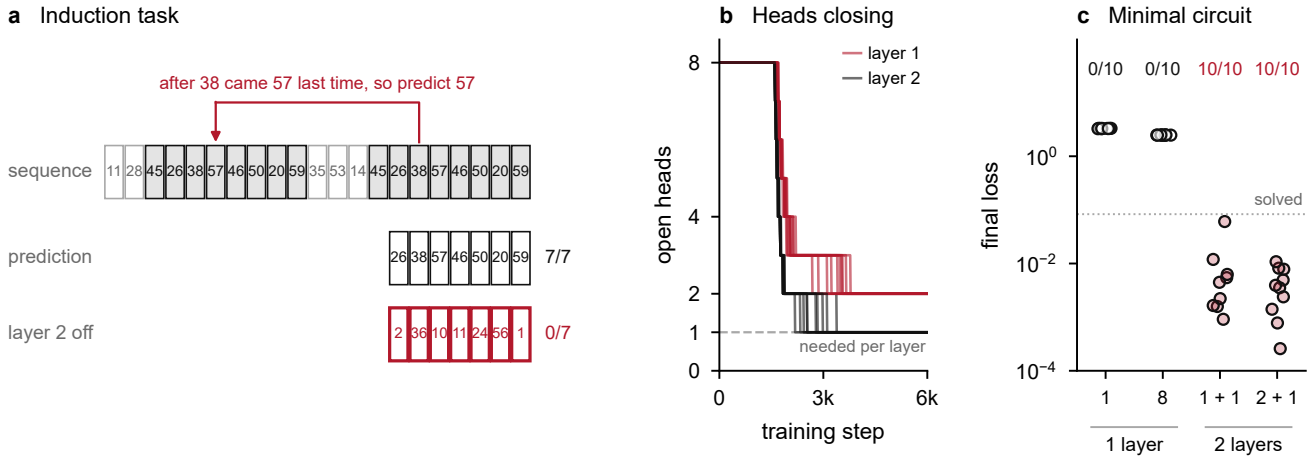


Figure 4: Induction. **(a)** The task: a random sequence repeats, and after token 38 the model should predict what followed 38 last time. With the kept layer-2 head switched off, the predictions are wrong. **(b)** Open heads per layer during training, 10 runs. Layer 2 closes to the one induction head, layer 1 to two previous-token heads. **(c)** Final loss of models trained from scratch with a fixed number of heads, 10 seeds each. One layer cannot solve the task with 1 or 8 heads; two layers solve it with $1 + 1$ and $2 + 1$, so the circuit the rule kept is minimal in layer 2 and one head above the minimum in layer 1.

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